

Homework 3

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1. Homework Setup

Problem 1

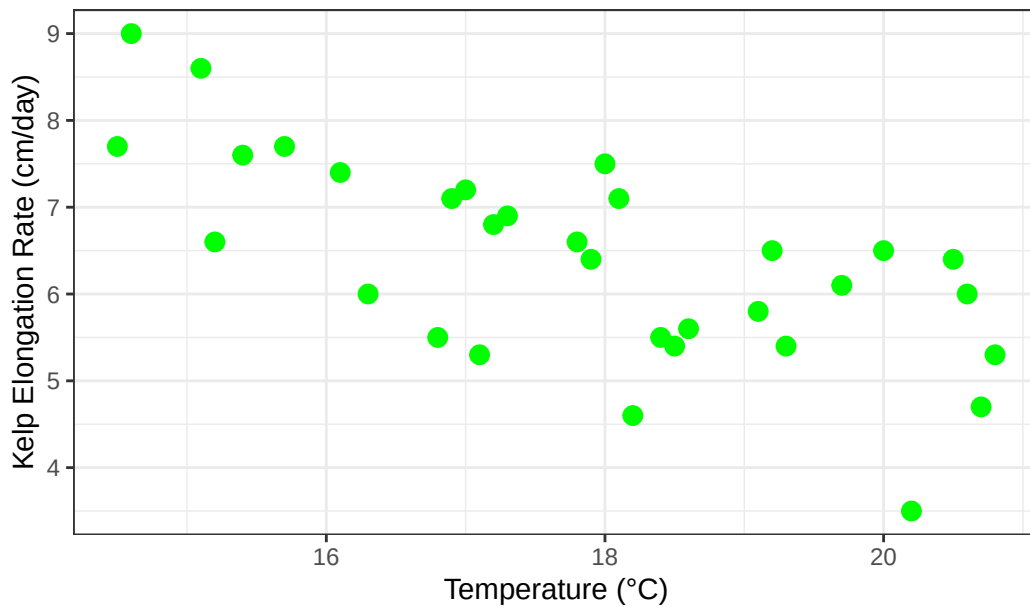
1. a

The two appropriate tests are the Pearson correlation and the Spearman rank correlation. Pearson correlation measures the strength of a linear relationship between two continuous variables and assumes the data are normally distributed, while Spearman rank correlation measures the strength of a monotonic relationship using ranked data and does not require normality.

1. b

```
#base layer: ggplot
ggplot(data = kelp,
       #x-axis: temperature
       #y-axis: kelp elongation
       mapping = aes(x = temp_c,
                     y = kelp_elong)) +
  #second layer: points for scatterplot
  geom_point(color = "green",
            size = 3) +
  #relabelling axes and title
  labs(x = "Temperature (°C)",
       y = "Kelp Elongation Rate (cm/day)",
       title = "As Temperature Increases, Kelp Frond Elongation Rate Decreases") +
  #changing theme from base
  theme_bw()
```

As Temperature Increases, Kelp Frond Elongation Rate Decreases



1. c

```
#running spearman rank correlation with temp and kelp elongation
cor.test(kelp$temp_c,
         kelp$kelp_elong,
         method = "spearman"
)
```

Spearman's rank correlation rho

```
data: kelp$temp_c and kelp$kelp_elong
S = 9216.1, p-value = 1.29e-05
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
-0.6891684
```

Spearman's $\rho = -0.69$

For the Spearman Rank Correlation, I checked for a monotonic relationship. I used the scatterplot in part 1b, which suggested a monotonic relationship, meaning that the assumption for Spearman's was met.

1. d

To evaluate the strength of the relationship between temperature and giant kelp frond elongation rate, I used a Spearman Rank Correlation, because it does not require the relationship between temperature and kelp elongation to be linear or the data to be normally distributed. We found a moderate to strong relationship between temperature and kelp frond elongation rate (Spearman's $r = -0.69$, $S = 9216.1$, $p < 0.001$, $\alpha = 0.05$).

1. e

The Spearman Rank Correlation test shows whether there is a consistent monotonic relationship between temperature and giant kelp frond elongation rate. The Spearman's $r = -0.69$ suggests a moderate to strong negative relationship between temperature and kelp frond elongation rate. Thus, kelp frond elongates at a higher rate at lower temperatures, and inversely at a lower rate at higher temperatures.

1. f

```
#running pearson rank correlation with temp and kelp elongation
cor.test(kelp$temp_c,
         kelp$kelp_elong,
         method = "pearson"
)
```

Pearson's product-moment correlation

```
data: kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
sample estimates:
      cor
-0.6877489
```

The Pearson Correlation test provided me with Pearson's $r = -0.69$, which is the same as the Spearman's r from Problem 1c. Pearson correlation tests for a linear relationship using raw values of temperature and kelp elongation rate, while Spearman correlation tests for a monotonic relationship using ranked values of the same variables. Both tests led me to the same result that there is a moderate to strong negative relationship between temperature and kelp frond elongation rate.